These types of functions are important in Ordinary differential equations (Chapter 10). Let us consider some examples.

Consider $F(x,y) = x^3 - 2y^3 + 5xy^2,\ (x,y) \in \mathbb{R}^2$. Then

$$F(\lambda x, \lambda y) = (\lambda x)^3 - 2(\lambda y)^3 + 5(\lambda x)(\lambda y)^2 = \lambda^3(x^3 - 2y^3 + 5xy^2)$$

and hence F is a homogeneous function of degree 3.

On the other hand,

$G(x,y) = e^{x^2} + 3y^2$ is not a homogeneous function because,

$$G(\lambda x, \lambda y) = e^{(\lambda x)^2} + 3(\lambda y)^2 \neq \lambda^p G(x,y)$$

for any $\lambda \neq 1$ and any p.

Example 8.21

Show that $F(x,y) = \dfrac{x^2 + 5xy - 10y^2}{3x + 7y}$ is a homogeneous function of degree 1.

Solution

We compute

$$F(\lambda x, \lambda y) = \frac{(\lambda x)^2 + 5(\lambda x)(\lambda y) - 10(\lambda y)^2}{3\lambda x + 7\lambda y} = \frac{\lambda^2}{\lambda}\left(\frac{x^2 + 5xy - 10y^2}{3x + 7y}\right) = \lambda F(x,y)$$

for all $\lambda \in \mathbb{R}$. So F is a homogeneous function of degree 1.

We state the following theorem of Leonard Euler on homogeneous functions.

Definition 8.13 (Euler)

Suppose that $A = \{(x,y) \mid a < b,\ c < y < d\} \subset \mathbb{R}^2$, $F : A \to \mathbb{R}^2$. If F is having continuous partial derivatives and homogeneous on A, with degree p, then

$$x\frac{\partial F}{\partial x}(x,y) + y\frac{\partial F}{\partial y}(x,y) = pF(x,y)\ \forall\ (x,y) \in A.$$

Suppose that $B = \{(x,y,z) \mid a < x < b,\ c < y < d,\ u < z < v\} \subset \mathbb{R}^3$, $F : B \to \mathbb{R}^3$. If F is having continuous partial derivatives and homogeneous on B, with degree p, then

$$x\frac{\partial F}{\partial x}(x,y,z) + y\frac{\partial F}{\partial y}(x,y,z) + z\frac{\partial F}{\partial z}(x,y,z) = pF(x,y,z)\ \forall\ (x,y,z) \in B.$$

We omit the proof. The above theorem is also true for any homogeneous function of n variables; and is useful in certain calculations involving first order partial derivatives.

Example 8.22

If $u = \sin^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, Show that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \dfrac{1}{2}\tan u$.

Solution

Note that the function u is not homogeneous. So we cannot apply Euler's Theorem for u.

However, note that $f(x,y) = \dfrac{x+y}{\sqrt{x}+\sqrt{y}} = \sin u$ is homogeneous; because

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